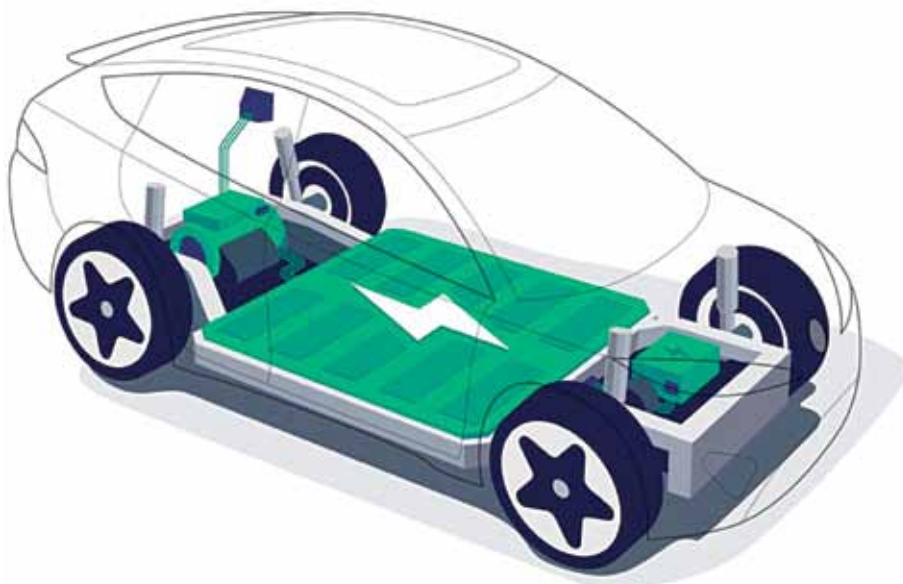




Better, safer batteries with material derived from trees



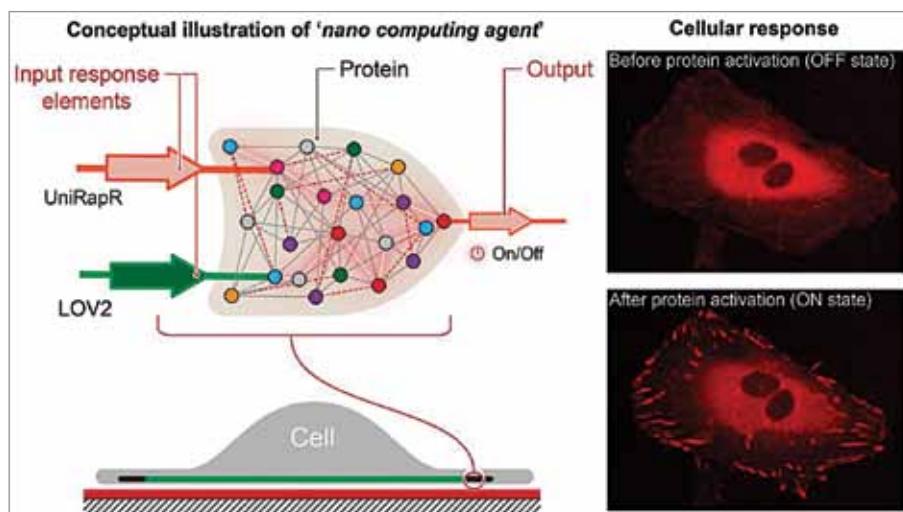
A material derived from trees could potentially replace liquid electrolytes in next-generation batteries (Credit: www.brown.edu)

In pursuit of batteries that deliver more power and operate more safely, researchers are looking for alternatives to replace the liquids commonly used in today's lithium-ion batteries. Now, a research team from Brown University and the University of Maryland has developed a new material for use in solid-state batteries that's derived from an unlikely source—trees. The team demonstrated a solid ion conductor that combined copper with cellulose nanofibrils—polymer tubes derived from wood. The paper-thin material can deliver an ion conductivity that is 10 to 100 times better than other polymer-ion conductors and could be used as either a solid battery electrolyte or as an ion-conducting binder for the cathode of an all-solid-state battery.



Nanoscale computer that controls human cell behaviour

The creation of nanoscale computers for use in precision healthcare has long been a dream of many scientists and healthcare providers. Now, for the first time, researchers at Penn State have produced a nanocomputing agent that can control the function of a particular protein that is involved in cell movement and cancer metastasis. The research paves the way for the construction of complex nanoscale computers for the prevention and treatment of cancer and other diseases, allowing deeper understanding of human biology and disease with the possibilities for introducing precision therapeutics.



Conceptual illustration of nano computing agent (Credit: www.psu.edu)